TfL Delay Intelligence System

Real-Time Anomaly Detection for Urban Transit

Modern ML Engineering Approach

Capstone Project | March 2026

Time-Aware Classification & Intelligent Monitoring

What We're Building vs What We're Not

✅ What This IS

- **Real-Time Risk Scoring System** - Streaming data ML with intelligent monitoring capabilities
- **Time-Aware Anomaly Detection** - Classification based on recent behavioral patterns and context
- **Production ML Engineering** - Real-time ingestion, feature engineering, and inference pipeline
- **Modern ML Pattern** - Similar to fraud detection and cybersecurity monitoring systems

❌ What This is NOT

- **Traditional Forecasting** - Not ARIMA models predicting future arrival times
- **Time Series Prediction** - Not LSTM networks forecasting next time steps
- **Simple Regression** - Not classical univariate time-series models
- **Academic Toy Problem** - Not Kaggle-style forecasting competitions

System Architecture: Three-Layer Intelligence

Layer 1: Data Collection & Storage

Real-time API monitoring with **ETL processes** for structured data ingestion. Efficient storage optimized for 7-day training horizons.

Tech Stack:

SQLite, Parquet, APIs

Layer 2: Feature Engineering & Intelligence

Time-aware features with rolling window statistics and seasonal conditioning. Multi-resolution spatiotemporal feature engineering.

Capabilities:

Rolling Windows, Seasonal Baselines

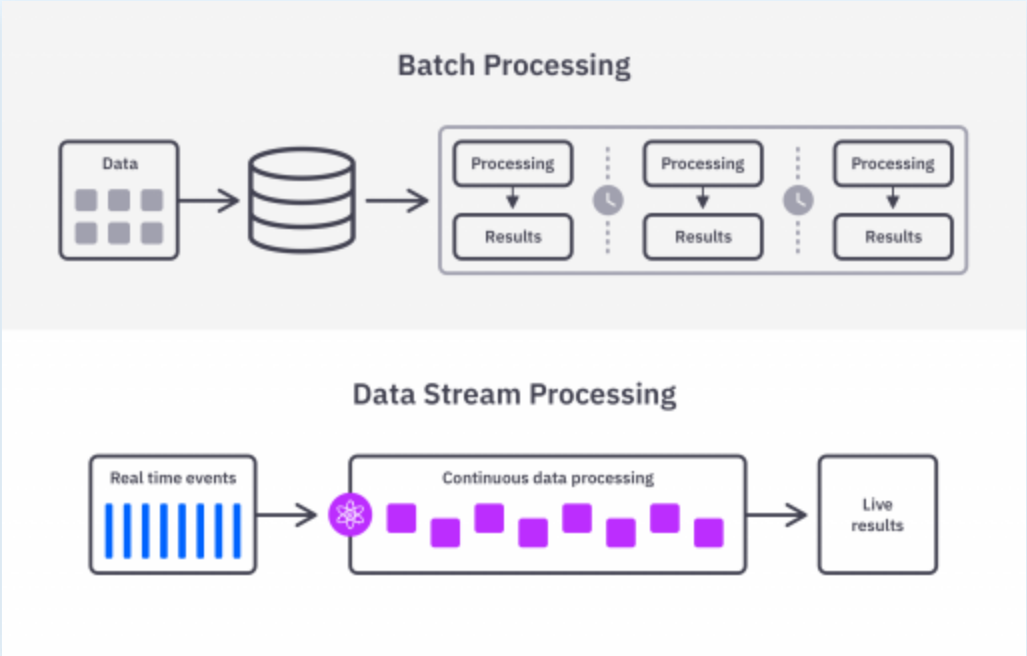
Layer 3: Classification & Risk Scoring

Binary classification using **logistic regression and gradient boosting**. PR-AUC evaluation with real-time risk scoring output.

Output:

Risk Scores, Anomaly Detection

Real-Time ML Pipeline



Modern streaming architecture for real-time ML inference

Time-Aware Processing Pipeline

Data Ingestion processes **real-time TfL API feeds** with sub-minute latency for immediate anomaly detection.

Feature Engineering creates **rolling window statistics** and seasonal baselines for temporal context.

Classification uses **gradient boosting** with PR-AUC optimization for imbalanced delay detection.

Risk Scoring outputs **real-time anomaly scores** for operational decision-making.

Key Insight: *This mirrors production fraud detection systems - classifying current behavior as normal/abnormal rather than predicting future values.*

Business Impact: £70-90M Annual Value

£70-90M

Annual Value in London Alone

Research shows that **accurate real-time transit information** saves passengers time valued at this amount annually. Our system directly addresses this multi-million pound problem.

Time Savings

Reduces passenger uncertainty and waiting times through better predictions

Operational Efficiency

Enables proactive resource allocation and disruption management

Sustainable Transport

Encourages public transport use by making it more reliable

Industry Applications & Real-World Parallels

Financial Services

Fraud Detection - Real-time transaction anomaly classification based on user behavior patterns

Credit Risk Scoring - Time-aware assessment of borrower risk using recent financial activity

Market Surveillance - Detecting unusual trading patterns and market manipulation attempts

Same Pattern:
Classify current transaction as normal/suspicious vs predicting next transaction amount

Cybersecurity

Threat Detection - Real-time network anomaly identification using behavioral baselines

User Behavior Analytics - Detecting compromised accounts through activity pattern analysis

System Monitoring - Identifying security incidents through log pattern anomalies

Same Pattern:
Classify current network activity as normal/malicious vs predicting future attack vectors

Industrial IoT

Predictive Maintenance - Equipment fault detection using sensor data pattern recognition

Process Monitoring - Manufacturing anomaly detection through real-time quality metrics

Safety Systems - Immediate hazard identification using environmental sensor networks

Same Pattern:
Classify current machine state as normal/faulty vs predicting exact failure time

Technical Differentiation: 2026 Market Relevance

Modern ML Engineering Capabilities

- **Real-Time Ingestion** **Sub-minute latency** processing with streaming data architecture
- **Safe DB Handling** **Production-grade** data validation and error handling
- **Time-Aware Validation** **Chronological splits** maintaining temporal dependencies
- **Explainability Layer** **SHAP integration** for transparent decision-making

> ***2026 Reality:*** *Recruiters value systems engineering over toy forecasting models. This project demonstrates* ***production ML patterns*** *used by top-tier companies.*

Time-Aware Classification vs Traditional Forecasting

Time-Aware Classification

Real-Time Anomaly Detection - Classifies current system behavior as normal or abnormal using ***rolling window context***

Temporal Memory - Uses ***seasonal conditioning*** and chronological validation for time-aware decisions

Production Pattern - Same approach used in fraud detection, cybersecurity monitoring, and industrial fault detection

Risk Scoring Output - Provides actionable ***probability scores*** for operational decision-making

PR-AUC Evaluation - Optimized for ***imbalanced datasets*** with proper performance metrics

Traditional Forecasting

Future Prediction - Attempts to predict exact arrival times using ***ARIMA/LSTM models***

Historical Patterns - Relies on ***univariate time series*** without rich contextual features

Academic Focus - Common in Kaggle competitions but less relevant for production ML systems

Limited Actionability - Predictions often lack ***immediate decision support*** needed for operations

RMSE/MAE Metrics - Uses regression metrics that don't align with ***business decision needs***

Career Opportunities & Target Companies

Target Career Paths & Companies

Smart Cities & Logistics

Transport for London, Google (Maps/Waymo), Uber, Citymapper

Direct experience with transit delay problems and urban mobility solutions

Quantitative Finance

Bloomberg, Jane Street, Two Sigma, Goldman Sachs

Mathematical similarities with market risk event detection and alpha prediction

Technology & AI

Microsoft, Amazon, Meta, NVIDIA

Real-time ML systems, streaming data processing, intelligent monitoring platforms

Energy & Infrastructure

Tesla, National Grid, Octopus Energy

Time-series skills apply to grid load forecasting and renewable energy prediction

Key Skills Demonstrated

Production ML Engineering Skills

- Real-Time Ingestion
- Time-Aware Validation
- Feature Engineering
- System Reliability
- Explainable AI
- Production Deployment

2026 Market Reality: The industry has shifted from "can you build a model?" to "can you build a system?" This project demonstrates both technical depth and engineering rigor valued by leading technology companies.

Production-Ready Engineering Standards

Enterprise-grade monitoring and reliability standards

Industry-Standard Engineering Practices

Data Safety Robust error handling and **data validation** with graceful degradation under system stress

System Reliability **>99% uptime** requirements with comprehensive monitoring and alerting

Scalability Design Efficient feature computation for **city-scale deployment** with resource optimization

Real-Time Constraints Sub-minute latency processing with **distributed capabilities** for large networks

Industry Alignment: *These engineering practices mirror the standards used at* ***FAANG companies*** *and demonstrate production ML system expertise valued in 2026.*

Future Extensions & Research Directions

Phase 2 Enhancements

Regression Components **Quantile prediction** for delay magnitude (P50/P90 estimates) with confidence intervals

Station-Level Forecasting **Multi-horizon prediction** capabilities for individual station performance

Network Propagation **Disruption modeling** across interconnected transport routes

Predictive Maintenance **Infrastructure monitoring** for proactive system health management

Implementation Timeline: *Phase 2 enhancements build on existing classification foundation, adding regression and network-level capabilities.*

Research Applications

Academic Contributions **Novel methodologies** for urban mobility modeling and time-aware feature engineering

Cross-Modal Integration **Multi-transport systems** combining bus, rail, and ride-sharing data

Evaluation Frameworks **Standardized metrics** for real-time classification system performance

Smart City Applications **Scalable deployment** patterns for global urban transportation networks

Research Impact: *Contributions to intelligent transportation systems and real-time ML methodology advancement.*

Key Takeaways

Modern ML Pattern

Real-time anomaly detection mirrors production systems at major tech companies - not academic forecasting

Quantified Business Impact

£70-90M annual value demonstrates understanding of real-world problem solving and economic impact

Production Engineering

Time-aware validation , safe DB handling, and explainability show industry- ready engineering practices

2026 Market Relevance

Aligns with current industry shift from **"can you build a model?"** to **"can you build a system?"**

TfL Delay Intelligence System

A sophisticated real-time ML system that demonstrates modern engineering practices, quantifiable business impact, and production-ready architecture.

Ready for Production ML Roles

Scalable & Future-Ready

Industry-Aligned Skills